

Problems Based upon Gradient, Divergence and Curl

PROBLEM 1. Find gradient of a scalar field $u = 3x^2y - y^3z^2$ at point $(1, -2, -1)$.

Solution: We know that gradient of scalar field u is

$$\begin{aligned}\nabla u &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (3x^2y - y^3z^2) \\ &= (6xy) \hat{i} + (3x^2 - 3y^2z^2) \hat{j} + (-2y^3z) \hat{k}\end{aligned}$$

At point $(1, -2, -1)$

$$\nabla u = -12 \hat{i} - 9 \hat{j} - 16 \hat{k} \quad \text{Ans.}$$

PROBLEM 2. Find a unit vector normal to the surface $x^3 + y^3 + 3xyz = 3$ at point $(1, 2, -1)$

Solution: Given surface $u = x^3 + y^3 + 3xyz = 3$

$$\therefore \nabla u = (3x^2 + 3y^2) \hat{i} + (3y^2 + 3x^2) \hat{j} + 3xy \hat{k}$$

At point $(1, 2, -1)$

$$\nabla u = -3 \hat{i} + 9 \hat{j} + 6 \hat{k}$$

The unit vector normal to the given surface u is given by

$$\hat{n} = \frac{\nabla u}{|\nabla u|} = \frac{-3 \hat{i} + 9 \hat{j} + 6 \hat{k}}{3\sqrt{14}} = \frac{1}{\sqrt{14}} (-\hat{i} + 3 \hat{j} + 2 \hat{k}) \quad \text{Ans.}$$

PROBLEM 3. Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $x^2 + y^2 - z = 3$ at point $(1, 1, 1)$.

Solution: The angle between given two surfaces will be equal to the angles between their normals. And normal to any surface is given by its gradient.

$$\text{Given surfaces} \quad u_1 = x^2 + y^2 + z^2 = 9$$

$$\text{and} \quad u_2 = x^2 + y^2 - z = 3$$

$$\therefore \nabla u_1 = 2x \hat{i} + 2y \hat{j} + 2z \hat{k}$$

$$\text{and} \quad \nabla u_2 = 2x \hat{i} + 2y \hat{j} - \hat{k}$$

At point $(1, 1, 1)$, gradients are:

$$\nabla u_1 = 2 \hat{i} + 2 \hat{j} + 2 \hat{k} \quad \dots(i)$$

$$\text{and} \quad \nabla u_2 = 2 \hat{i} + 2 \hat{j} - \hat{k} \quad \dots(ii)$$

Let θ be the angle between two surfaces, then

$$\cos \theta = \frac{\vec{n}_1 \cdot \vec{n}_2}{\left| \vec{n}_1 \right| \left| \vec{n}_2 \right|}$$

where $\vec{n}_1$ and $\vec{n}_2$ are the normals to the given surfaces and are respectively represented by eqns. (i) and (ii)

$$\therefore \cos \theta = \frac{4+4-2}{\sqrt{12} \cdot \sqrt{9}} = \frac{6}{\sqrt{12} \cdot 3} = \frac{2}{\sqrt{12}}$$

$$\therefore \theta = \cos^{-1}\left(\frac{2}{\sqrt{12}}\right) \quad \text{Ans.}$$

PROBLEM 4. Find the divergence and curl of a vector field

$$\vec{F} = (xyz)\hat{i} + (3x^2y)\hat{j} + (xz^2 - y^2z)\hat{k} \text{ at point } (1, 1, 1).$$

Solution: Given

$$\vec{F} = (xyz)\hat{i} + (3x^2y)\hat{j} + (xz^2 - y^2z)\hat{k}$$

$\therefore$ Divergence

$$\vec{F} = \vec{\nabla} \cdot \vec{F} = yz + 3x^2 + 2xz - y^2$$

At point (1, 1, 1), we have

$$\vec{\nabla} \cdot \vec{F} = 1 + 3 + 2 - 1 = 5 \quad \text{Ans.}$$

Now

$$\text{curl } \vec{F} = \vec{\nabla} \times \vec{F} = -2yz\hat{i} + (xy - z^2)\hat{j} + (6xy - x^2)\hat{k}$$

At point (1, 1, 1), we have

$$\vec{\nabla} \times \vec{F} = -2\hat{i} + 0\hat{j} + 5\hat{k} = -2\hat{i} + 5\hat{k} \quad \text{Ans.}$$

PROBLEM 5. Show that the vector field $\vec{F} = (x^2 + xy^2)\hat{i} + (y^2 + x^2y)\hat{j}$ is irrotational.

Solution: We know that if curl of a vector field is zero, field is irrotational one

$$\begin{aligned} \therefore \vec{\nabla} \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x^2 + xy^2) & (y^2 + x^2y) & 0 \end{vmatrix} \\ &= (0 - 0)\hat{i} + (0 - 0)\hat{j} + (2xy - 2xy)\hat{k} = 0 \end{aligned}$$

$\therefore \vec{F}$ is irrotational.

PROBLEM 6. Find the constants a, b, c such that the vector field

$$\vec{F} = (x + 2y + az)\hat{i} + (bx - 3y - 2)\hat{j} + (4x + cy + 2z)\hat{k} \text{ is irrotational.}$$

Solution: We know that the vector field $\vec{F}$ is irrotational if its curl is zero, i.e., $\vec{\nabla} \times \vec{F} = 0$

$$\therefore \vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x + 2y + az) & (bx - 3y - 2) & (4x + cy + 2z) \end{vmatrix} = 0$$

or

$$\begin{aligned} \hat{i} \left[\frac{\partial}{\partial y} (4x - cy + 2z) - \frac{\partial}{\partial z} (6x - 3y - 2) \right] \\ + \hat{j} \left[\frac{\partial}{\partial z} (x + 2y + az) - \frac{\partial}{\partial x} (4x - cy + 2z) \right] \\ + \hat{k} \left[\frac{\partial}{\partial x} (6x - 3y - 2) - \frac{\partial}{\partial y} (x + 2y + az) \right] = 0 \end{aligned}$$

or

$$\hat{i} (c + 1) + \hat{j} (a - 4) + \hat{k} (b - 2) = 0$$

$\Rightarrow$

$$c + 1 = 0; a - 4 = 0; b - 2 = 0$$

or

$$a = 4, b = 2, c = -1 \quad \text{Ans.}$$

PROBLEM 7. Show that the vector field $\vec{F}$ is solenoidal.

Given

$$\vec{F} = (x + 3y)\hat{i} + (y - 3z)\hat{j} + (x - 2z)\hat{k}.$$

Solution: We know that for a vector field $\vec{F}$ to be solenoidal, its divergence should be zero, i.e.,

$$\vec{\nabla} \cdot \vec{F} = 0$$

$$\begin{aligned} \therefore \vec{\nabla} \cdot \vec{F} &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) ((x + 3y)\hat{i} + (y - 3z)\hat{j} + (x - 2z)\hat{k}) \\ &= (1 + 1 - 2) = 0 \end{aligned}$$

Hence given vector field $\vec{F}$ is solenoidal.

PROBLEM 8. A point charge of 12 C is located at the centre of a cube of side 5 m. Calculate the electric flux through each of the surface.

Solution: Since the charge is located at the centre of the cube, electric flux through each of the surface is same. Therefore, flux through any surface is $\frac{1}{6}$ th of the total flux.

$\therefore$ Total flux from a charge of 12 C is 12 C.

Hence flux through one of the surfaces of the cube is $\frac{12}{6}$ C, i.e., 2 C.

Ans.

PROBLEM 9. A circular plane sheet of radius 1 m is placed in an electric field of 200 N/C making an angle 60° with the field. Calculate the electric flux through the sheet.

Solution: We know that the electric flux through a surface of cross-section S is

$$\phi_E = \vec{D} \cdot \vec{S} = \epsilon_0 \vec{E} \cdot \vec{S} = \epsilon_0 E S \cos \theta$$

where θ is the angle between $\vec{E}$ and $\vec{S}$

Given

$$E = 200 \text{ N/C} = \pi \times 1^2 = \pi \text{ m}^2 \text{ and } \cos 60^\circ = \frac{1}{2}$$

Also

$$\epsilon_0 = 8.85 \times 10^{12} \text{ C}^2/\text{N/m}^2$$

$\therefore$

$$\phi_E = \frac{1}{2} \times 8.85 \times 10^{12} \times 200 \times \pi = 2780 \times 10^{12} \text{ C}$$

Ans.

PROBLEM 10. Given that intensity of electric field $\vec{E}$ in a certain charge distribution is given by $\vec{E} = r^{n-1}\hat{r}$, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$. Find the volume charge density.

Solution: According to Gauss's law

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

Since

$$\vec{E} = r^{n-1}\hat{r} = r^{n-1} \frac{\vec{r}}{r} = r^{n-2}\vec{r} = (x^2 + y^2 + z^2)^{(n-2)/2} (x\hat{i} + y\hat{j} + z\hat{k})$$

Also

$$\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$$

$\therefore$

$$E_x = (x^2 + y^2 + z^2)^{(n-2)/2} x$$

which gives,

$$\begin{aligned} \frac{\partial E_x}{\partial x} &= \left(\frac{n-2}{2} \right) (x^2 + y^2 + z^2)^{\frac{n-2}{2}-1} 2x \cdot x + (x^2 + y^2 + z^2)^{\frac{n-2}{2}} \\ &= (n-2)(x^2 + y^2 + z^2)^{(n-4)/2} x^2 + (x^2 + y^2 + z^2)^{(n-2)/2} \\ &= (n-2)r^{n-4}x^2 + r^{n-2} \end{aligned}$$

Similarly,

$$\frac{\partial E_y}{\partial y} = (n-2)r^{n-4}y^2 + r^{n-2}$$

and

$$\frac{\partial E_z}{\partial z} = (n-2)r^{n-4}z^2 + r^{n-2}$$

$\therefore$

$$\begin{aligned} \vec{\nabla} \cdot \vec{E} &= \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = (n-2)r^{n-4}(x^2 + y^2 + z^2) + 3r^{n-2} \\ &= (n-2)r^{n-4}r^2 + 3r^{n-2} = r^{n-2}(n-2+3) = (n+1)r^{n-2} \end{aligned}$$

$\therefore$

$$\rho = \epsilon_0 \vec{\nabla} \cdot \vec{E} = \epsilon_0 (n+1)r^{n-2}$$

Hence **Proved**.

PROBLEM 11. A capacitor with two parallel 1 cm $\times$ 2 cm each receives 2.25 V potential difference between the electrodes. (a) How far apart must the plates be to produce a charge density of 10^{-7} C/m²? (b) How many electrons accumulate on the -ve plate? No dielectric is there.

Solution: (a) We know that, $D = \epsilon_0 E = \epsilon_0 \frac{V}{d}$

or

$$d = \frac{\epsilon_0 V}{D} = \frac{8.85 \times 10^{-12} \times 2.25}{10^{-7}} = 0.0002 \text{ m}$$

(b) Also the number of electrons are given by

$$n = \frac{\text{Area of plate} \times \text{charge density}}{\text{Charge on electron}} = \frac{1 \times 2 \times 10^{-11}}{1.6 \times 10^{-19}} = 125 \times 10^6 \text{ electrons} \quad \text{Ans.}$$

PROBLEM 12. The electric potential V due to a charge in the surrounding space at any point x meters from the charge is given by the relation $V = 8x + 3x^2$ volts. Find the electric field at point 1.5 m from the charge. Consider medium as air.

Solution: Given potential at a point, $V = 8x + 3x^2$

$$\therefore \quad \text{At } x = 1.5 \text{ m}$$

$$V = 8 \times 1.5 + 3(1.5)^2 = 18.75 \text{ volts}$$

Let Q be the charges due to which at 1.5 m the potential is 18.75 volts. By definition for air being medium, potential at a point R m from the charge is $V = \frac{Q}{4\pi\epsilon_0 R}$

Given

$$R = 1.5 \text{ m}$$

$\Rightarrow$

$$Q = 3.125 \times 10^{-9} \text{ coulomb}$$

Now the intensity due to Q charge at a point x metre from the charge is given by

$$E = \frac{Q}{4\pi\epsilon_0 x^2} = \frac{3.125 \times 10^{-9}}{4\pi\epsilon_0 (1.5)^2} = 12.5 \text{ N/C}$$

Ans.

PROBLEM 13. A charge Q is distributed over two concentric hollow spheres of radii r_1 and r_2 ($r_2 > r_1$) such that the surface densities are equal. Find the potential at the common centre.

Solution: Let the charge be distributed such that inner sphere has charge q_1 and outer sphere has charge q_2 , i.e., $Q = q_1 + q_2$... (i)

Given surface densities are equal, i.e.,

$$\frac{q_1}{4\pi r_1^2} = \frac{q_2}{4\pi r_2^2}$$

or

$$q_1 r_2^2 = q_2 r_1^2$$

... (ii)

Solving for q_1 and q_2 , we have from eqns. (i) and (ii),

$$q_1 = \frac{Q r_1^2}{r_1^2 + r_2^2} \text{ and } q_2 = \frac{Q r_2^2}{r_1^2 + r_2^2}$$

Potential due to these charges at the centre will be equal to the sum of potential due to q_2 on the outer sphere and q_1 on the inner sphere, i.e.,

$$\begin{aligned} V &= \frac{q_1}{4\pi\epsilon_0 r_1} + \frac{q_2}{4\pi\epsilon_0 r_2} = \frac{1}{4\pi\epsilon_0} \left[\frac{Q r_1^2}{r_1 (r_1^2 + r_2^2)} + \frac{Q r_2^2}{r_2 (r_1^2 + r_2^2)} \right] \\ &= \frac{Q}{4\pi\epsilon_0} \left[\frac{r_1}{(r_1^2 + r_2^2)} + \frac{r_2}{(r_1^2 + r_2^2)} \right] = \frac{Q}{4\pi\epsilon_0} \left(\frac{r_1 + r_2}{r_1^2 + r_2^2} \right) \end{aligned}$$

Ans.

PROBLEM 14. For what value of a , the vector field

$$F = (ax + 3y)\hat{i} + (y - 2z)\hat{j} + (x - 2z)\hat{k} \text{ is solenoidal.}$$

Solution: We know that a vector field is solenoidal if its divergence is zero, i.e., $\vec{\nabla} \cdot \vec{F} = 0$

$$\therefore \quad \vec{\nabla} \cdot \vec{F} = \frac{\partial}{\partial x}(ax + 3y) + \frac{\partial}{\partial y}(y - 2z) + \frac{\partial}{\partial z}(x - 2z) = a + 1 - 2$$

Since

$$\vec{\nabla} \cdot \vec{F} = 0$$

$\therefore$

$$a = 1 \text{ Ans.}$$

PROBLEM 15. Find the unit vector normal to the surface $xy^3 z^2 = 10$ at a point $(-1, -1, 2)$.

Solution: A vector normal to the given surface is obtained by computing its gradient i.e. ∇S where S is the surface.

Given

$$S = xy^3 z^2 = 10$$

$\therefore$

$$\nabla S = y^3 z^2 \hat{i} + 3xy^2 z^2 \hat{j} + 2xy^3 z \hat{k}$$

At point $(-1, -1, 2)$, we have

$$\nabla S = -4\hat{i} - 12\hat{j} + 4\hat{k}$$

$\therefore$ Unit vector normal to the surface

$$= \frac{\nabla S}{|\nabla S|} = \frac{-1}{\sqrt{11}} (\hat{i} + 3\hat{j} - \hat{k}) \quad \text{Ans.}$$

PROBLEM 16. In an electric field the electric potential is given by $U(x, y, z) = (4x^2 + 3y^2 + 9z^2)^{-1/2}$. Calculate the electric field at point (111).

Solution: The electric field and the potential U are related as;

$$\vec{E} = -\vec{\nabla} U$$

$\therefore$

$$\begin{aligned} \vec{E} &= -\vec{\nabla} U(x, y, z) = -\vec{\nabla} [4x^2 + 3y^2 + 9z^2]^{-1/2} \\ &= -\left[\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right] [4x^2 + 3y^2 + 9z^2]^{-1/2} \\ &= \frac{1}{2} (4x^2 + 3y^2 + 9z^2)^{-1/2} (\hat{i} 8x + \hat{j} 6y + \hat{k} 18z) \end{aligned}$$

$\therefore$

$$\begin{aligned} \vec{E}(111) &= \frac{1}{2} (4 + 3 + 9)^{-3/2} (\hat{i} 8 + \hat{j} 6 + \hat{k} 18) \\ &= \frac{(\hat{i} 8 + \hat{j} 6 + \hat{k} 18)}{2.16^{3/2}} = \frac{\hat{i} 8 + \hat{j} 6 + \hat{k} 18}{2.4^3} = \frac{4\hat{i} + 3\hat{j} + 9\hat{k}}{64} \quad \text{Ans.} \end{aligned}$$

Problems on Polarization

PROBLEM 17. A parallel plate capacitor consists of free space and two dielectrics of relative dielectric constants $\epsilon_r = 2$ and $\epsilon_r = 3$ respectively. If the plates of the capacitor have a charge density of $\rho_s = 5 \times 10^{-6} \text{ C/m}^2$, find the electric displacement, electric field and electric polarization in the three regions.

Solution: The given capacitor consists of three regions (i) free space, (ii) dielectric with $\epsilon_r = 2$ and (iii) dielectric with $\epsilon_r = 3$. Now consider the first region, i.e., free space

(a) The electric displacement $D_1 = \rho_s = 5 \times 10^{-6} \text{ C/m}^2$

(b) The electric field $E_1 = \frac{\rho_s}{\epsilon_0} = \frac{5 \times 10^{-6}}{8.854 \times 10^{-12}} = 0.56 \times 10^6 \text{ V/m}$

(c) The electric polarizations $P_1 = \text{zero}$

In region second with dielectric $\epsilon_r = 2$

(a) $D_2 = \rho_s = 5 \times 10^{-6} \text{ C/m}^2$

(b) $E_2 = \frac{D_2}{\epsilon_0 \epsilon_r} = \frac{5 \times 10^{-6}}{8.854 \times 10^{-12} \times 2} = 0.28 \times 10^6 \text{ V/m}$

$$(c) P_2 = \epsilon_0 (\epsilon_r - 1) E_2 = 8.855 \times 10^{-12} (1) 0.28 \times 10^6 = 2.48 \times 10^{-6} \text{ C/m}^2$$

In third region with dielectric $\epsilon_r = 3$

$$(a) D_3 = \rho_s = 5 \times 10^{-6} \text{ C/m}^2$$

$$(b) E_3 = \frac{D_3}{\epsilon_0 \epsilon_r} = \frac{5 \times 10^{-6}}{8.854 \times 10^{-12} \times 3} = 0.188 \times 10^6 \text{ V/m}$$

$$(c) P_3 = \epsilon (\epsilon_r - 1) E_3 = 8.854 \times 10^{-12} (2) 0.188 \times 10^6 = 3.3 \times 10^{-5} \text{ C/m}^2$$

On comparing three regions, we conclude that the electric displacement remains same, the electric field E goes on decreasing and the electric polarization goes on increasing along free space to dielectric with $\epsilon_r = 2$ to dielectric with $\epsilon_r = 3$.

PROBLEM 18. An electric field of 10^5 V/m is applied on a sample of neon at NTP. Calculate the dipole moment induced in each atom. Given dielectric constant for neon = 1.000134. Also find the atomic polarizability of neon gas.

Solution: We know that

$$P = \epsilon_0 (\epsilon_r - 1) E$$

and

$$P = np$$

Comparing above two relations, we have

$$p = \frac{\epsilon_0 (\epsilon_r - 1) E}{n}$$

Given $\epsilon_r = 1.000134$, $E = 10^5 \text{ V/m}$

We can calculate the number of atoms of neon gas at NTP as follows

$$n = \frac{6.023 \times 10^{26}}{22.4}$$

$$\therefore p = \frac{8.85 \times 10^{-12} (1.000134 - 1) 10^5 \times 22.4}{6.023 \times 10^{26}} = 3.8 \times 10^{-36} \text{ Cm Ans.}$$

Also, the atomic polarizability

$$\alpha = \frac{p}{E} = \frac{3.8 \times 10^{-36}}{10^5} = 3.8 \times 10^{-41} \text{ Cm}^2/\text{V Ans.}$$

PROBLEM 19. The relative permitting of Ar at 0°C and one atmosphere is 1.000435. Calculate the polarizability of the atom?

Solution: The number of atom of Ar at NTP are $\frac{6 \times 10^{26}}{22.4} = 2.6 \times 10^{25}$

We know that

$$P = \epsilon_0 (\epsilon_r - 1) E \text{ and } P = Np$$

$$\therefore p = \frac{\epsilon_0 (\epsilon_r - 1) E}{N} \text{ or } \frac{p}{E} = \frac{\epsilon_0 (\epsilon_r - 1)}{N}$$

$$\Rightarrow \alpha = \frac{\epsilon_0 (\epsilon_r - 1)}{N}$$

where α is the polarizability

Given $\epsilon_r = 1.000435$

$$\epsilon_r = \frac{8.85 \times 10^{-12} (1.000435 - 1)}{2.6 \times 10^{25}} = 1.43 \times 10^{-40} \text{ F m}^2.$$

PROBLEM 20. An atom of oxygen an being polarized produces a dipole moment of 0.5×10^{-22} c.m. If the distance of the centre of negative charge cloud from nucleus be 4×10^{-15} cm, calculate the polarizability of the oxygen atom.

Solution: We know that is equilibrium position, the Lorentz force and the Coulomb interaction are equal

i.e.,
$$(8e) E = \frac{(8e)(8e)}{4\pi\epsilon_0 r^2}$$

where charge on oxygen atom is $8e$ and the distance between negative charge cloud and the nucleus is r

$$\therefore E = \frac{8e}{4\pi\epsilon_0 r^2}$$

Given

$$r = 4 \times 10^{-17} \text{ m}$$

$\therefore$

$$E = \frac{8 \times 1.6 \times 10^{-19}}{4\pi \times 8.85 \times 10^{-12} \times (4 \times 10^{-17})^2} = 2.6 \times 10^{24} \text{ V/M.}$$

Also, we know that dipole moment p is given by

$$p = \alpha E$$

or

$$\alpha = \frac{p}{E}$$

Given

$$p = 0.5 \times 10^{-22} \text{ C-m}$$

$\therefore$

$$\alpha = \frac{0.5 \times 10^{-22}}{2.6 \times 10^{24}} = 1.9 \times 10^{-47} \text{ Fm}^2. \quad \text{Ans.}$$

PROBLEM 21. Determine the percentage of ionic polarizability in the NaCl crystal which has refractive index 1.5 and the static dielectric constant 5.6. Given $\epsilon_r = \mu^2$.

Solution: We know that $\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{\epsilon_r - 1}{\epsilon_r + 2}$

Since the given sample is NaCl crystal, so we can have ionic as well as electronic polarizabilities i.e., $\alpha = \alpha_i + \alpha_e$

$$\therefore \frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{n(\alpha_i + \alpha_e)}{3\epsilon_0}$$

Given $\epsilon_r = \mu^2$

$\therefore$ The %age of ionic polarizability

$$\begin{aligned} &= \left(\frac{\alpha_i}{\alpha_i + \alpha_e} \right) \times 100 = \left(1 - \frac{\alpha_e}{\alpha_i + \alpha_e} \right) \times 100 \\ &= \left[1 - \frac{(\epsilon_r + 2)(\mu^2 - 1)}{(\epsilon_r + 1)(\mu^2 + 2)} \right] \times 100 \end{aligned}$$

($\because$ electronic polarization occurs only in optical region)

= 51.3% **Ans.**

PROBLEM 23. Show that potential function

$U = x^2 + y^2 - 2z^2$ satisfies the Laplace's equation.

Solution: Laplace's equation is:

$$\nabla^2 U = 0 \text{ or } \frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2} = 0$$

Here

$$\frac{\partial U}{\partial x} = 2x \text{ and } \frac{\partial^2 U}{\partial x^2} = 2$$

$$\frac{\partial U}{\partial y} = 2y \text{ and } \frac{\partial^2 U}{\partial y^2} = 2$$

$$\frac{\partial U}{\partial z} = -4z \text{ and } \frac{\partial^2 U}{\partial z^2} = -4.$$

Hence,
$$\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2} = 2 + 2 - 4 = 0$$

It satisfies the Laplace's equation.

PROBLEM 24. A current of 10A flows through a conductor of cross-section 1 mm². If density of the charge carriers be 10²¹ cm⁻³, calculate the drift velocity of the electrons. Charge on electron is 1.6 × 10⁻¹⁹ C.

Solution:

$$J = nev = \frac{1}{S}$$

∴

$$I = nevS$$

Here,

$$I = 10A$$

$$S = 1 \text{ mm}^2 = 10^{-6} \text{ m}^2$$

$$n = 10^{21} \text{ cm}^{-3} = 10^{27} \text{ m}^{-3}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

Hence

$$\begin{aligned} v &= \frac{1}{neS} = \frac{10A}{10^{27} \text{ m}^{-3} \times 1.6 \times 10^{-19} \text{ C} \times 10^{-6} \text{ m}^2} \\ &= \frac{10 \text{ Cs}^{-1}}{10^2 \times 1.6 \text{ Cm}^{-1}} = \frac{1}{16} \text{ ms}^{-1} = 0.0625 \text{ ms}^{-1} \end{aligned}$$

PROBLEM 25. Using Maxwell's equation $\text{div } D = \rho$, prove that $F = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^3} r$

Solution: Given $\text{div } D = \rho$

Integrating over a sphere of radius r , we get

$$\int_S \text{div } D \, dv = \int_V \rho \, dv = q, \text{ the total charge}$$

Changing volume integral into surface integral

$$\int_V D \cdot ds = q$$

or

$$D_4 \pi r^2 = q$$

or

$$\epsilon_0 E_4 \pi r^2 = q$$

or

$$E = \frac{q}{4\pi\epsilon_0 r^2}$$

Force on a test charge q_0 will be

$$F = q_0 E = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^3} r$$

Hence Proved.

Also we know that moving charge produces a magnetic field and is given by

$$B = \frac{\mu_0 2I}{4\pi R} = \frac{\mu_0 I}{2\pi R}$$

$$\therefore B = \frac{4\pi \times 10^{-7} \times 1.1 \times 10^{-3}}{2 \times \pi \times 5.1 \times 10^{-11}} = 14 \text{ Wb/m}^2$$

PROBLEM 27. Given that $\vec{B} = \vec{\nabla} \times \vec{A}$ and $\vec{\nabla} \cdot \vec{A} = 0$. Then prove that

$$\vec{\nabla}^2 \vec{A} = -\mu_0 \vec{J}$$

Solution:

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

or

$$\vec{\nabla} \times \vec{B} = \vec{\nabla} \times (\vec{\nabla} \times \vec{A})$$

But

$$\vec{A} \times \vec{B} \times \vec{C} = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C}$$

$\therefore$

$$\vec{\nabla} \times \vec{B} = (\vec{\nabla} \cdot \vec{A})\vec{\nabla} - (\vec{\nabla} \cdot \vec{\nabla})\vec{A}$$

Given

$$= \vec{\nabla} \cdot \vec{A} = 0$$

Therefore,

$$\vec{\nabla} \times \vec{B} = -(\vec{\nabla} \cdot \vec{\nabla})\vec{A} = -\vec{\nabla}^2 \vec{A}$$

But

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

$\therefore$

$$-\vec{\nabla}^2 \vec{A} = \mu_0 \vec{J}$$

Hence,

$$-\vec{\nabla}^2 \vec{A} = \mu_0 \vec{J} \quad \dots(i)$$

This is the Poisson's equation for $\vec{A}$.

If $J = 0$ then the eqn. (i) reduces to:

$$\nabla^2 \vec{A} = 0 \quad \dots(ii)$$

Which is the Laplace's equation for $\vec{A}$.